THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A right triangle has hypotenuse 29 cm and one leg 20 cm. Its area is: 1

(a) 420 cm² **(b)** 210 cm² **(c)** 290 cm² **(d)** 304 cm²

2. The diagonal of a square is 14 cm. Its area is: 1

(a) 98 cm² **(b)** 49 cm² **(c)** 196 cm² **(d)** 28 cm²

3. An L-shaped figure is made by removing a 12 cm by 9 cm rectangle from a 30 cm by 20 cm rectangle. Its area is: 1

(a) 600 cm² **(b)** 492 cm² **(c)** 708 cm² **(d)** 198 cm²

4. A parallelogram has base 24 cm and height 9 cm. If the base is changed to 18 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 11 **(b)** 12 **(c)** 15 **(d)** 198

5. A rhombus has side 10 cm and one diagonal 12 cm. Its area is: 1

(a) 40 cm² **(b)** 120 cm² **(c)** 96 cm² **(d)** 48 cm²

6. A path 1 m wide runs around the outside of a 24 m by 16 m rectangular garden. The area of the path is: 1

(a) 4 m² **(b)** 468 m² **(c)** 84 m² **(d)** 80 m²

7. A right triangle has hypotenuse 41 cm and one leg 9 cm. Its area is: 1

(a) 184 cm² **(b)** 180 cm² **(c)** 820 cm² **(d)** 360 cm²

8. The diagonal of a square is 16 cm. Its area is: 1

(a) 256 cm² **(b)** 64 cm² **(c)** 128 cm² **(d)** 32 cm²

9. An L-shaped figure is made by removing a 9 cm by 5 cm rectangle from a 24 cm by 16 cm rectangle. Its area is: 1

(a) 429 cm² **(b)** 339 cm² **(c)** 384 cm² **(d)** 165 cm²

10. A parallelogram has base 30 cm and height 8 cm. If the base is changed to 20 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 10 **(b)** 12 **(c)** 220 **(d)** 15

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A rhombus has side 30 cm and one diagonal 36 cm. Find its other diagonal and area. 2

12. A square garden has a diagonal of 22 m. Find its area and side length (leave the side as a surd). 2

13. A rectangle has perimeter 120 cm and length 42 cm. Find its breadth and area. 2

14. A rhombus has side 34 cm and one diagonal 32 cm. Find its other diagonal and area. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangular park is 45 m by 28 m. A path 2 m wide runs inside along its boundary. Find the area of the path. 3

16. A parallelogram has base 36 cm and height 8 cm. Another side of 24 cm has a corresponding height; find that height and the area. 3

17. A right triangle has legs 12 cm and 5 cm. Find its hypotenuse and area, and the length of the altitude drawn to the hypotenuse. 3

OR

A rectangular park is 60 m by 40 m. A path 4 m wide runs inside along its boundary. Find the area of the path. 3

18. A parallelogram has base 28 cm and height 15 cm. Another side of 20 cm has a corresponding height; find that height and the area. 3

OR

A right triangle has legs 20 cm and 15 cm. Find its hypotenuse and area, and the length of the altitude drawn to the hypotenuse. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A figure is made of a rectangle 30 cm by 16 cm with a right triangle of base 30 cm and height 12 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. 5

OR

The parallel sides of a trapezium are 36 cm and 24 cm and its area is 240 cm². Find the distance between the parallel sides. A parallelogram of base 24 cm (the shorter parallel side) and the same height is cut out, leaving a triangle; find the area of the parallelogram and of the triangle left. 5

20. A figure is made of a rectangle 16 cm by 9 cm with a right triangle of base 16 cm and height 6 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. 5

OR

The parallel sides of a trapezium are 40 cm and 20 cm and its area is 270 cm². Find the distance between the parallel sides. A parallelogram of base 20 cm (the shorter parallel side) and the same height is cut out, leaving a triangle; find the area of the parallelogram and of the triangle left. 5

END OF QUESTION PAPER — Total: 40 Marks